

Linear-Attention Mixer Bake-Off — Run 1

run_019fe0f9-1bbd-702c-b141-6d58e128bda6 · 18/18 cells, 0 failures · 6 operators × 3 seeds · 390M params, 1,144 steps = 599,785,472 tokens/cell (TPP 1.5) · reservoir-dolma2-v1 · 8xA100-40GB · \$790.47 declared ceiling · 2026-08-08

Every arm is identical except the **operator in decoder slots {6, 11}**. All six keep the same six global-attention layers (2,5,8,10,12,14) — mandatory, because RoPE buffers are emitted only for Attention blocks, so dropping attention drops positional encoding and the loss gap stops being a mixer effect. All arms parameter-matched within ±0.05% of 390,135,552 by FFN-width respond.

1 · WHAT EACH ARM ACTUALLY IS

Arm (mechanism)	Config	What differs from the reference	Params Δ	FLOP/token
KDA, no conv activation KDA_NOACT	conv_activation=None gated_conv=False	Shipped KDA with the SiLU removed from the three short convolutions; no gate. Exists as the isolating control for the gated-conv arm.	−10,080	2,473,388,544
KDA + LIV gated convolution KDA_GCONV	gated_conv=True gate_structure=depthwise	LFM2/LIV-style depthwise gates on all three short convolutions. Depthwise not lowrank: lowrank would cost +2,359,296 params (12× tolerance) and forfeit seed pairing.	+2,208	2,473,425,408
KDA, as shipped (reference) KDA_BASE	fla_chunk_kda gated_conv=False	The shipped Kimi Delta Attention operator — per-channel forget gate, scalar beta in (0,1), fla's fused chunked kernel. Every contrast subtracts this.	−10,080	2,473,388,544
KDA-Householder, negative eigenvalues, R=2 KDA_R2	num_householder=2 allow_neg_eigval=True	Two Householder (DeltaProduct) factors per token, beta widened to (0,2) so (I − βkk ^T) can be a true reflection. Custom non-chunked Triton kernel.	+6,304	2,457,037,312
KDA-Householder, negative eigenvalues, R=1 KDA_R1	num_householder=1 allow_neg_eigval=True	Arity control. One factor per token, beta in (0,2). Mathematically the same function as KDA_BASE — identical params, FLOPs and state_dict — differing only in that it runs the custom kernel. Isolates reflection regime from arity.	−10,080	2,473,388,544
Gated DeltaNet-2 GDN2	expand_v=1.0 allow_neg_eigval=False	Decouples the erase and write gates that KDA drives from one scalar beta, into two independent channel-wise gates; per-channel decay through a low-rank bottleneck.	+22,688	2,512,375,232

1B · PROS AND CONS, PER OPERATOR — EVERY CLAIM IS A MEASURED NUMBER FROM THIS RUN

Arm	Pros	Cons
KDA, no conv activation KDA_NOACT	+ Best CE of all six (3.0400), −0.0114 vs reference. + Most reproducible arm in the run — sd 0.00258, 4.2× tighter than the reference. + Free: same params (−10,080), same peak <i>and</i> reserved memory (9.153 / 10.721 GiB), 0.998× throughput. + Simplest change in the set — one config field, no new tensors, no new kernel. + Runs fla's fused chunk_kda, so it inherits the numerics gate that already passed.	− CE advantage does not clear the MDE (CI [−0.0597, +0.0370] spans zero) — a bound, not a win. − Removing the SiLU is an unexplained gain: no mechanism is proposed for why it should help, so it may be noise that happened to land favourably. − Only tested at TPP 1.5; an activation may matter more with more tokens. − Exists as a control, so it has had no design attention as a candidate.
KDA + LIV gated convolution KDA_GCONV	+ Second-best CE (3.0451). + Parameter-matched at +2,208 (0.0006% of the model) — quality is not bought with capacity. + Gate verified <i>live</i> on sm_80: 360/360 convolutions fused, gate gradients 1.4e-05 against a 1.0e-07 floor, gates moved 0.020 off neutral. + Bit-identical to the plain operator at init (0.000e+00), so the ablation is exact.	− Against its correct control the post-gate is +0.0051 nats — nominally worse, CI [−0.0312, +0.0414], p = 0.765. − Highest peak memory of any arm: 9.434 GiB, +3.1%. − Slowest of the four fla-kernel arms: 0.979×, MFU 40.7%. − Second-widest seed spread (sd 0.01678) — 6.5× the no-activation arm. − The depthwise pre-gate is algebraically a SiLU, so only 1 real dof/channel is new; the headline −0.0062 vs the reference is an artefact of the wrong comparator. − Lowrank variant is unusable: +2,359,296 params (12× tolerance) and it forfeits seed pairing.
KDA, as shipped (reference) KDA_BASE	+ Fastest arm measured (419,288 tok/s) and joint-lowest memory (9.153 / 10.721 GiB). + Highest MFU tier (41.5%) alongside the other fla-kernel arms. + Best-supported path: fla's fused chunked kernel, already numerics-gated, already run on sm_80 at exit 0. + No CE contrast in this run beats it by a resolvable margin — it is not shown to be worse	− Mid-table on CE (3.0514); three arms have lower point estimates, none resolvably so. − Seed spread sd 0.01075, 4.2× wider than the no-activation arm — the reference is itself not especially stable. − Its per-channel forget gate keeps β in (0,1), so the update is a contraction and no reflection mechanism is available. − Paper variant, not the K3 variant: no lower-bounded gate, low-rank output gate.

than anything.
+ Zero integration risk; it is the operator already in the tree.

KDA-Householder, negative eigenvalues, R=2
KDA_R2

+ Tight seed spread (sd 0.00642), 1.7× better than the reference — the mechanism trains consistently.
+ Parameter-matched at +6,304 despite a mixer 47% larger (6,608,976 vs 4,487,248), respend in FFN width.
+ Does *fewer* FLOPs/token than the reference (2,457,037,312 vs 2,473,388,544), so its slowness is provably kernel inefficiency and not extra work.
+ The marginal cost of the second Householder factor is only **10.9%** — the arity itself is cheap.
+ $\beta \in (0,2)$ makes $(I - \beta k k^T)$ a true reflection: the state-tracking mechanism is genuinely present.

– **Worst efficiency in the run: 0.705× throughput, MFU 29.1%** — a 29% tax.
– CE is +0.0067 vs the reference: **no quality gain at any measurable level**.
– Reserved memory **13.391 GiB, +24.9%** — the O(B·T·H·K·V) fp32 backward workspace, invisible in the peak figure and an OOM risk at longer context.
– Custom non-chunked Triton kernel: not covered by fla’s numerics gate, no CP/TP support, torch.compile-disabled.
– Would need the chunking rewrite (~4–7 eng-days) before it is shippable.

KDA-Householder, negative eigenvalues, R=1
KDA_R1

+ Scientifically the most valuable arm in the run: it is the **only** way the Householder result could be split into kernel cost vs arity cost.
+ Parameter- and FLOP-identical to the reference, which doubles as an equivalence check on the implementation.
+ $\beta \in (0,2)$ reflection regime available at nominally KDA-equivalent maths.

– **Never ship this**. Same function as the reference, 0.779× the speed (MFU 32.4% vs 41.5%) — the 28.6% is pure kernel overhead for zero mathematical difference.
– **Least reproducible arm by a wide margin: sd 0.04516**, 17× the no-activation arm, on identical mathematics. One seed landed at 3.1267 against siblings at 3.0542/3.0439.
– That single arm’s variance **inflated the pooled σ from 0.01025 to 0.02042** and dissolved GDN2’s otherwise-significant contrast — it damaged the whole run’s power.
– Worst CE of the KDA family (3.0749).
– Reserved memory 13.260 GiB (+23.7%) for no mechanism the reference lacks.

Gated DeltaNet-2
GDN2

+ Second-most reproducible (sd 0.00404) and highest MFU in the run (42.0%).
+ Throughput essentially free at 0.994× — it uses fla’s fused chunk_gdn2, not a custom kernel.
+ Genuinely different mechanism: erase and write gates decoupled from one scalar β , plus per-channel decay through a low-rank bottleneck.
+ Best-tested implementation in the tree — its kernel-contract test transcribes the paper’s recurrence independently and reduces to KDA when the gates are tied.

– **Worst CE of all six (3.0884), +0.0371 vs the reference** — the largest effect measured in the run, in the wrong direction, and replicated across all three seeds (3.0838/3.0910/3.0905).
– The only arm whose contrast is significant under *any* test (Welch/T3 CI [+0.0262, +0.0479], $p = 0.0005$), so this is the least likely result to be noise.
– Costs +2.0% peak memory (9.333 GiB) for that.
– Largest parameter surplus to respend (+22,688 after solve_widths).
– Not param-matched to anything at its defaults; expand_v and allow_neg_eigval both had to be pinned by hand.

Reproducibility is read from the per-arm seed sd: an arm whose three seeds disagree by more than the effect being hunted cannot support a production decision, whatever its mean. **Reserved** memory is quoted alongside peak for the Householder arms because their fp32 backward workspace shows up there and not in the peak-allocated figure.

2 · RESULTS — HELD-OUT CROSS-ENTROPY (PRIMARY) AND COST (CO-PRIMARY)

Arm	n	val_ce (nats)	sd	per-seed values		Δ vs ref	Dunnett 95% CI	clears MDE?	tok/s	xref	peak GiB	xref	MFU
KDA, no conv activation	3	3.0400	0.00258	3.0392	/ 3.0380 / 3.0429	−0.01135	[−0.0597, +0.0370]	no	418,364	0.998	9.153	1.000	41.5%
KDA + LIV gated convolution	3	3.0451	0.01678	3.0321	/ 3.0392 / 3.0641	−0.00624	[−0.0546, +0.0421]	no	410,690	0.979	9.434	1.031	40.7%
KDA, as shipped (reference)	3	3.0514	0.01075	3.0421	/ 3.0632 / 3.0488	—	—	—	419,288	1.000	9.153	1.000	41.5%
KDA-Householder, neg. eigenvalues, R=2	3	3.0581	0.00642	3.0567	/ 3.0651 / 3.0524	+0.00669	[−0.0417, +0.0551]	no	295,638	0.705	9.361	1.023	29.1%
KDA-Householder, neg. eigenvalues, R=1	3	3.0749	0.04516	3.0542	/ 3.1267 / 3.0439	+0.02358	[−0.0248, +0.0719]	no	326,513	0.779	9.216	1.007	32.4%
Gated DeltaNet-2	3	3.0884	0.00404	3.0838	/ 3.0910 / 3.0905	+0.03705	[−0.0113, +0.0854]	no	416,894	0.994	9.333	1.020	42.0%

Throughput is steady-state, post-50-step warmup, summed over 8 devices. Peak memory is a true per-step running max; its sd is **exactly 0.000 within every arm**, so those ratios are mechanism, not seed noise.

3 · STATISTICS

Quantity	Value	Reading
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Pooled within-arm $\hat{\sigma}$	0.02042 nats, df = 12	χ^2 interval [0.01464, 0.03370] — a factor-2.3 bracket, the pre-registered target precision. Size any future run from this , not from the 0.0019/0.0105 literature pair.
Minimum detectable effect	0.0636 nats	80% power, exact non-central t. The literature gap between these mixer families is 0.010–0.030 nats — below the MDE .
One-way ANOVA	F(5,12) = 2.489, p = 0.091	Arms not distinguishable at $\alpha = 0.05$ under pooled variance.
Dunnett vs reference	crit 2.9013, k = 5	No contrast clears the MDE . CE is not resolved at n = 3.
Variance homogeneity	Levene p = 0.484 Bartlett p = 0.0059	They disagree. R=1's sd is 17× the no-activation arm's, so pooling lets one unstable arm widen every CI. Under the pre-registered unequal-variance fallback (Welch F = 53.29, p = 0.0005) GDN2's T3 CI is [+0.0262, +0.0479] — excluding zero . Both are reported; neither is settled.
Gate contrast gated conv – no-activation KDA	+0.00511 nats [−0.0312, +0.0414], p = 0.765	The only attributable test of the gated conv's post-gate . The depthwise pre-gate is algebraically a SiLU (2·sigmoid(a·u)·u $\equiv$ (2/a)·silu(a·u), exact to 8.9e-16), so comparing to the shipped reference instead moves three things at once — and reads −0.0062, the opposite sign . A bound, not a benefit.

4 · RECOMMENDATION

Use KDA as shipped, or KDA with the convolution activation removed. Basis: **throughput and memory, because cross-entropy is unresolved**. The four fla-kernel arms sit within 2% on speed and 0.05 nats on CE; the two custom-kernel Householder arms cost 22% and 29% for no measurable quality.

- **The Householder finding is about the kernel, not the mechanism**. R=1 is mathematically identical to the shipped operator — same params, same 2,473,388,544 FLOP/token — yet runs at 0.779× with MFU 32.4% against 41.5%. So **kernel overhead alone is 28.6%**, while the arity step R=1→R=2 costs a further **10.9%**. If R>1 is wanted, chunk the kernel; never ship KDA-Householder at R=1, since the shipped operator is the same model 28.6% faster.
- **R=1 also carries sd 0.04516** — 4× any other arm, on identical mathematics. Suspected kernel-induced instability; n=3 with one outlier cannot prove it. Worth checking against the backend="torch" reference path.
- **The gated convolution is a bound, not a benefit**: +0.005 nats against its correct control, for +3.1% peak memory and −2.1% throughput.
- **GDN2 is the largest effect measured** (+0.037 nats, worse) and the only one significant under any test. Directional, not settled — and a quality-at-fixed-params result at TPP 1.5, not a verdict on the implementation.

5 · CAVEATS THAT TRAVEL WITH EVERY CE NUMBER ABOVE

- **TPP = 1.5**, below the 3–20 academic cluster (budget cut 1,907→1,144 steps to fit the window). Architecture effects measured at low token budgets **systematically overstate** — measured in-tree, GDN's edge halved 0.0103 nats @1B → 0.0059 @15B. **Treat every CE figure as an upper bound on the production effect**.
- **Throughput and peak memory are budget-independent and fully valid**. They do not inflate the way CE does, which is why they carry the recommendation.
- **A null is a bound, not equivalence**. No margin was pre-declared, so no TOST is available; a non-significant contrast licenses only "smaller than the CI half-width".
- A 0.6B-token run structurally cannot see effects that emerge late in training, or any long-horizon recall advantage. That is a limitation, not a null.
- Secondary endpoint (sliced_eval) is null throughout: it needs --slice-mask-uri, which this command does not set. Declared, not quietly skipped.

Per-cell integrity, all 18: step-0 loss 11.71 inside the ln(100,352) = 11.5164 ± 0.5 band · val_tokens_present == declared exactly (975,077,376 over all 39 held-out shards) · parameter count matched ARM_L0_DELTA · seed pair matched the frozen schedule · 0 hard errors. Data seeds 210007/220014/230021 shared across arms (identical token order); init seeds unique per arm. Source sha 9425fea; results committed 562dbd7 under docs/mixer-bakeoff/run1/.